Jheronimus Academy of Data Science

Thesis project: format for the Research Proposal

Name **Friso Veldhorst** **Student number 1510290**

Preliminary title of your thesis

Rethinking Shape: Developing a Novel Circular Object Detection Method using Deep Learning

Main supervisor: _Pieter Blok_______________________________________

A brief description of your thesis proposal

Provide an abstract of 500 words maximum

In modern agriculture, accurate detection and size estimation of fruits and vegetables play an important role in automating crop monitoring, yield prediction, and selective harvesting. Existing deep learning approaches for object detection, such as YOLO and RF-DETR , rely on rectangular bounding boxes to localize objects in images. While effective for general-purpose detection tasks, these rectangular representations fail to capture the shape of many natural circular-shaped objects, including fruits and vegetables. A practical challenge when applying rectangular boxes on circular-shaped objects is that the output shape will contain redundant background pixels and that it doesn't mimic its real shape leading to inaccurate localization and size estimation, especially in the presence of occlusion

This thesis proposes a novel approach for circular object detection by rethinking the geometric representation used in deep learning architectures. The central research question guiding this thesis is: *Can the prediction of naturally circular objects with a circle-shaped object detection model be improved compared to a standard rectangular-shaped object detection model?*

To address this question, the project aims to design and evaluate a neural network that predicts circular parameters, which are the centre coordinates and the radius of objects, instead of the traditional bounding boxes. By integrating a circular representation into the prediction head of a detection model, the model is expected to achieve more precise detection and more reliable size estimates, as well as decrease the computational power needed as the number of parameters is

lowered.
The proposed method will build upon insights from recent studies such as *Circle Representation for Medical Object Detection* (Nguyen, et al., 2021), which demonstrated the advantages of circle-based predictions in the medical imaging domain, and *Image-Based Size Estimation of Broccoli Heads under Varying Degrees of Occlusion* (Blok, van Henten, van Evert, & Kootstra, 2021), which highlighted the importance of accurate object size estimation under challenging visual conditions. Adapting these concepts to the agricultural context, the research will involve (1) constructing a circular detection head compatible for the state-of-the-art RF-DETR object detection model (Robinson, Robicheaux, & Popov, 2025), (2) defining a suitable loss function for circle regression, and (3) benchmarking the performance against conventional box-based models using the broccoli's dataset of Blok, van Henten, van Evert, & Kootstra.
Experimental evaluation will focus on metrics such as detection performance expressed by mean average precision (mAP) and intersection over union (IoU). The detection performance measures the ability of a model to detect the object in an image. Through these metrics, the study also aims to assess robustness in scenarios with high object density and partial occlusion, conditions frequently encountered in real-world crop cultivation.

Beyond the technical innovation, this research carries significant entrepreneurial and societal impact. By improving object detection precision even marginally, by a few percentage points, agricultural robots and harvesting systems can achieve higher efficiency, and faster return on investment. Moreover, more accurate fruit size estimation can support better yield forecasting and reduce food waste by minimizing errors in automated sorting and harvesting processes.
Ultimately, this thesis seeks to contribute to the field of geometrically aware object detection by demonstrating how direct circle prediction can enhance performance in shape-specific domains, such as cell detection in the medical sector and bubble detection in the petrochemical industry. The expected outcomes include a functional prototype model, an open-source implementation, and a benchmark comparison highlighting the benefits of circular detection for agricultural applications to contribute towards smarter, more sustainable food production systems.

I am doing the master thesis based on (please select the appropriate option):

- My own company, namely: ____
- External company or organization, namely: ____
- Research project @ JADS, namely: <u>Rethinking Shape: Developing a Novel Circular Object Detection Method using Deep Learning</u>

Problem statement and research question

a) Describe your field of study and the existing body of knowledge. What lacunae still exist in this area? What has been ignored so far? What question exists in scholarly literature, in theory, or in practice that points to the need for meaningful understanding and deliberate investigation?

The field of study is deep learning and more specifically image-based object detection. Within the field of object detection, I want to look specifically at circular agricultural products. In the agricultural domain, precise object detection is essential for tasks like fruit counting, size estimation, and yield prediction. Traditional rectangular bounding boxes with models like YOLO and RF-DETR (Robinson, Robicheaux, & Popov, 2025) fail to capture the natural shape of agricultural objects, such as fruits and vegetables, which are typically round. This leads to inefficiencies, as the rectangular bounding box may either include unnecessary background or miss parts of the object. To address this challenge, we want to explore and implement a new neural network architecture for circular object detection. By directly predicting circles with the deep learning network, one will be able to improve the accuracy of object localization of fruits and vegetables, especially in dense environments where objects overlap. This work could revolutionize crop monitoring, providing more precise data for tasks such as yield prediction and autonomous harvest, ultimately leading to a more efficient and sustainable food production.

This new model will be tested on a broccoli dataset (Blok, van Henten, van Evert, & Kootstra, 2021). Assuming this is a success, this model could be further extended and tested to other circle-shaped natural object domains, like other crops, bubbles or blood particles. Right now, there are already algorithms which draw a bounded circle around the object they are detecting. However, these algorithms all first detect their object using a conventional object-detection model (i.e. R-CNN), after which they do some sort of manipulation to draw a circle.

In the case of *Image-based size estimation of broccoli heads under varying degrees of occlusion* (Blok, van Henten, van Evert, & Kootstra, 2021), the ORCNN algorithm first predicts a bounding box and then predicts 2 masks inside the bounding box, one for the visible region and one for the amodal region (amodal is the combined visible and non-visible region belonging to an object of interest). This combined output demonstrated an improvement of the size estimate of the broccoli heads as an input for selective harvesting, although it led to a complex neural network architecture that need much input data for efficient optimization.

BubCNN: Bubble detection using Faster RCNN and shape regression network (Haas, Schubert, Eickhoff, & Pfeifer, 2020) has the same idea. It is a two-stage deep learning system for detecting and measuring gas bubbles in multiphase flows.
It first locates bubbles in images by predicting bounding boxes around the bubbles using a faster R-CNN detector. After that, a shape regression CNN approximates each bubble's shape as an ellipse (i.e., center, major/minor axes, and rotation). This hybrid design integrates the localization efficiency of region-based detectors with the geometric precision of a regression network — allowing automatic, generalizable bubble analysis under diverse experimental conditions.

As for *Circle representation for Medical Object Detection* (Nguyen, et al., 2021), CircleNet is an anchor-free deep learning model designed to detect ball-shaped biomedical objects (like glomeruli or nuclei) more efficiently than traditional box-based detectors. Instead of predicting rectangular bounding boxes (x, y, width, height), CircleNet predicts circular regions described by: The center point (cx, cy) and the radius r. This design makes CircleNet rotation-invariant, simpler (fewer degrees of freedom), and better aligned with naturally round biological objects. In essence, this paper discusses almost the technology as we want to experiment with, except it builds on the framework of CenterNet and is focused on the biomedical sector, whereas we want to build on the framework of the state-of-the-art RF-DETR transformer algorithm, which has different neural network components compared to CenterNet.

One of the research gaps in this area is to adapt the RF-DETR model to immediately draw bounding circles around the object rather than first draw a bounding box and making a mask within this bounding box. We hypothesize that this could improve the object detection of crops under occlusion due to the neural network being optimized to the natural shape of the crop. We think this could be the case due to the fact that with this adaptation, the step of making a circle based on the mask of the object gets eliminated as the circle is already predicted by the neural network architecture in the altered detection head.

Please state your target scientific discipline that you intend to contribute to (e.g., Data modeling and analysis methods, Deep learning, Data engineering, HTI and recommender systems, Entrepreneurship, Strategy, Legal/ethics, Social networks, Regulations and institutions, etc.):

Deep learning, Computer science, Object detection, Python

Do you intend to pursue a publication based on your thesis and write up the thesis report in the form of a scientific article consistent with standards of a reputable journal/ISI-rated conference from this discipline? (we expect you to agree on this with your main supervisor and (later) 2nd assessor explicitly before answering this question).

Yes/No.

In case Yes: what is the target journal/ ISI-rated conference? Artificial intelligence in agriculture

b) What core question needs to be answered to contribute to achieving your research (i.e., resolving the issue identified in the problem statement)? What sub-questions need to be addressed to answer the core question?

The core question that needs to be answered is: In what way does adapting object detection models to a circular shape prior influence the prediction of naturally round objects. This means that we want to implement a direct bounding circle to the most state-of-the-art model: RF-DETR, and measure whether this improves the performance of the standard rectangular-based RF-DETR model.

This immediately leads us to a few sub-questions:
1) Which architectural components of the RF-DETR model need to be modified to enable direct prediction of circular object parameters (center coordinates and radius) instead of rectangular bounding boxes?
2) How should the loss function be reformulated to effectively optimize circular predictions, ensuring accurate localization and size estimation while maintaining training stability within the RF-DETR framework?
3) How can the RF-DETR data loading pipeline be adjusted to support a custom dataset annotated with circular parameters, and what preprocessing steps are required to ensure correct label formatting and efficient training?
4) To what extent does the adapted circular RF-DETR model improve detection accuracy and size estimation performance compared to its rectangular baseline on the broccoli dataset?

The innovative character of the proposed project

What is the scientific and practical significance of your thesis? Does it contain an original contribution to the field of existing knowledge, and what is this contribution exactly? (consider using the patterns in scientific contributions per discipline described in the Study Guide and the slides of the Information Session)

The practical significance of this thesis would be that an improved model would increases the revenue generated for both the sellers of agricultural robots, as a better model would increase the sales of their robots. It could also increase the revenue for farmers, as this potential improvement could decrease the amount of crops the robots miss, thus getting a higher yield of crops. Whilst the thesis would in essence be an incremental model improvement, we think the consequences of this thesis could be that costs are decreased and revenue is increased. This would mean this thesis falls between 2 and 3 on the maturity scale.

The scientific significance of this thesis is that there is a gap within the current research field. This thesis could fill this gap, by either proving or disproving the hypothesis.

Proposition, hypotheses, and concepts

What is the central proposition? What are the core assumptions and working hypotheses? (in case you have any of the latter). What are the main concepts you intend to use, and how do you plan to operationalize them?

The central proposition is that through optimizing a model by using bounding circles, considering the natural circle-shape of crops, the estimation of the scale and location of crops under (partial) occlusion would be more accurate. Therefore, our hypothesis is:

With a circular-shaped object detection model, the prediction of naturally circular objects will be improved due to a tailored detection and optimization task within the neural network.

We hypothesize this adapted model only improves the performance for naturally round objects, so that is what the scope is limited to. In particular, for this thesis the scope is detection of circular-shaped broccoli heads.

Research design

Describe the research design you plan to use. Make sure it fits the project goal and research question(s). Describe your variables/features and indicate the methods you plan to use and analyses you plan to conduct. What data do you intend to gather, how are you planning to do this, and do you have access to these data? How do you plan to safeguard the reliability and validity of your research?

The research design involves first constructing a circular detection head compatible with the object detection model RF-DETR and also adjusting the dataloader such that the right targets are loaded in. After that, a suitable loss function for circle regression should be made to measure the performance of the adapted RF-DETR made in step 1. Last but not least, this adapted model should be compared to the performance of conventional box-based model using the broccoli dataset. This allows us to validate whether the adapted model is an improvement or not.

This broccoli dataset is open-source, provided by Blok et al. (Blok, van Henten, van Evert, & Kootstra, 2021). It is available on the following github: https://github.com/pieterblok/sizecnn.

Entrepreneurship

Please describe the business and/or societal impact your thesis project will have. Which of the three business impact maturity levels do you target? (1. Novel solution for decreasing costs in the existing business, 2. Novel solution for increasing revenues in the existing business, 3. New revenue generation/new business development/new product/new service).

How do you intend to do so? Which experiments/tests do you want to run to substantiate your impact? Which customers, companies, or organizations will you focus on in this validation? (could also be outside the organization in which you do the master thesis)

The increasing costs of labour within the agricultural sector is leading farmers to look for ways to automate their farms. The major way to do this is by using robots, but these robots can only be successful when they are able to detect crops even when these are planted close to each other or suffer from occlusion (this means that crops are partly covered by leaves). Right now, this occlusion leads to harvesting losses, resulting in food waste, food shortage and a lowered trust by farmers. Improving this aspect of object detection in the robots could lead to less food waste, food shortage more trust by farmers which increases the commercial value of the robot.
This makes the entrepreneurial impact large as a few percent better detection will lead to a faster ROI, resulting in more investment from the farmers. Furthermore, it helps preventing food waste, which is a nice beneficial societal impact. This means the outcome of this thesis has a business impact maturity level of 2.

Name of the arranged 2nd assessor and in case help needed in finding one, suggestions for the expertise of the 2nd assessor:

Danielle Sent

References

Blok, P. M., van Henten, E. J., van Evert, F. K., & Kootstra, G. (2021). Image-based size estimation of broccoli heads under varying degrees of occlusion. *Biosystems Engineering*, 213-233.

Haas, T., Schubert, C., Eickhoff, M., & Pfeifer, H. (2020). BubCNN: Bubble detection using Faster RCNN and shape regression network. *Chemical Engineering Science*.

Nguyen, E. H., Yang, H., Deng, R., Lu, Y., Zhu, Z., Roland, J. T., . . . Huo, Y. (2021). Circle Representation for Medical Object Detection. *Institute of Electrical and Electronics Engineers*, 746-754.

Robinson, I., Robicheaux, P., & Popov, M. (2025, October 28). *RF-DETR [Computer software]*. Opgehaald van Github: https://github.com/roboflow/rf-detr